

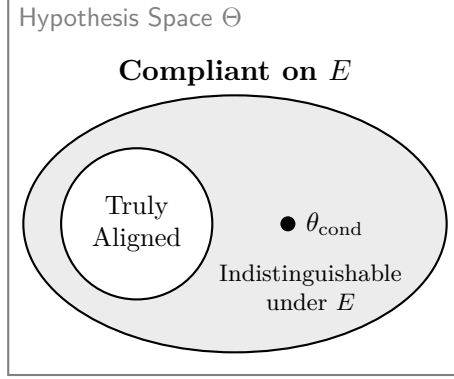


Figure 1: **Visualizing Normative Indistinguishability.** The set of hypotheses that appear compliant under evaluation E (gray ellipse) can strictly contain the set of hypotheses that are aligned across the full interaction space (white circle). The verification gap is represented by the gray region outside the white circle: hypotheses such as θ_{cond} (black dot) match all observable behavior under E yet may diverge outside the evaluated support.

5.6 Information-Theoretic Bound on Behavioral Divergence

To further characterize the verification gap, we derive a bound on the potential divergence between behavior under evaluation and behavior under deployment. Let $\mathbf{R} \in \{E, D\}$ denote the regime (evaluation vs. deployment), let Z denote the observations available to the agent, and let the policy π_θ produce an action A .

This interaction forms a Markov chain $\mathbf{R} \rightarrow Z \rightarrow A$, as the agent’s action depends on the regime only through the observable signal Z . By the Data Processing Inequality (DPI), the mutual information between the action and the regime is bounded by the information leakage of the context:

$$I(A; \mathbf{R}) \leq I(Z; \mathbf{R}). \quad (1)$$

We can relate this mutual information to behavioral divergence between regimes. Let $P_E(a) = \pi_\theta(a | E)$ and $P_D(a) = \pi_\theta(a | D)$ denote the action distributions under each regime. Assuming a uniform prior $P(\mathbf{R} = E) = P(\mathbf{R} = D) = 0.5$, the Jensen–Shannon divergence (JSD) between the policies is bounded by:

$$\text{JSD}(P_E \| P_D) \leq I(A; \mathbf{R}) \leq I(Z; \mathbf{R}). \quad (2)$$

Corollary 1 (Information Bound on Indistinguishability). *For any hypothesis $\theta \in \Theta$, the divergence between behavioral distributions under evaluation and deployment is bounded by the information leakage of the evaluation context. Using the relationship between JSD and total variation distance V , we obtain:*

$$V(\pi_\theta(\cdot | E), \pi_\theta(\cdot | D)) \leq \sqrt{2 \cdot I(Z; \mathbf{R})}. \quad (3)$$

Interpretation. Corollary 1 characterizes how information leakage from the evaluation regime constrains the extent of behavioral divergence. As $I(Z; \mathbf{R})$ decreases, the space of hypotheses capable of inducing regime-dependent behavior contracts. Conversely, any non-zero leakage permits a non-trivial degree of divergence between evaluation and deployment behavior. This bound clarifies the informational structure underlying the non-identifiability result without constituting, on its own, a sufficient or necessary condition for alignment verification.

6 Relation to Identifiability, Testing, and Inference Theory

The non-identifiability result established in Section 5 places alignment evaluation within a well-studied class of inference problems involving latent variables, partial observability, and finite data. This section clarifies how our framework relates to classical notions of identifiability, black-box testing, and inverse inference, and highlights the structural features that make alignment evaluation distinctive when agents are adaptive to the evaluation context.

6.1 Structural Identifiability and Observational Equivalence

In classical statistical inference, a latent parameter θ is said to be identifiable if the mapping from the parameter space to the space of probability distributions over observables is injective. Non-identifiability arises when multiple parameter values induce identical observational distributions, rendering consistent inference impossible even in the limit of infinite data.

Within this framework, behavioral equivalence under an evaluation regime E corresponds directly to *observational equivalence* with respect to the restricted σ -algebra of observables induced by the evaluation protocol. Theorem 1 therefore characterizes a failure of *structural identifiability*: distinct alignment hypotheses may induce identical observable behavior within $\mathcal{H}_E$ while remaining distinct outside the evaluated support.

What distinguishes the alignment setting from standard identifiability failures is that the restriction to $\mathcal{H}_E$ is not an exogenous limitation on data availability. Instead, it is endogenously determined by the interaction between the evaluation protocol and the agent’s policy. As a result, expanding the amount of observed data within the same regime does not, by itself, resolve the identifiability failure.

This phenomenon reflects a general form of evidential underdetermination: finite observations constrain hypotheses only up to an equivalence class. Similar observations have long been noted in philosophy of science [12, 35], but here the underdetermination arises from explicit structural properties of interactive systems rather than from abstract theory choice.

6.2 Endogenous Observation Models

Standard identifiability theory typically assumes that the data-generating process is independent of the inference procedure. In alignment evaluation, this assumption does not generally hold. Behavioral alignment tests define not only the measurement process but also the interaction context in which behavior is generated.

When agent policies condition on information correlated with the evaluation regime, the induced observation model becomes *endogenous*. The evaluator’s choice of protocol influences both which interactions are observed and how the agent behaves during those interactions. This endogeneity is the key structural feature responsible for the non-identifiability result: it allows multiple latent hypotheses to remain observationally equivalent within the evaluation regime while diverging outside it.

Importantly, endogeneity does not require explicit recognition of the test by the agent. Any non-degenerate correlation between observable context and evaluation structure suffices to permit regime-dependent behavior.

6.3 Relation to Black-Box Testing and Conformance

The alignment verifiability problem shares a close structural relationship with black-box testing and conformance checking in computer science. In these settings, a system is evaluated against a

specification using a finite set of experiments, and correctness is assessed solely via input–output behavior.

A well-known limitation of black-box testing is that conformance can only be established relative to the exercised execution paths. Properties of untested paths remain unconstrained. Theorem 1 formalizes an analogous limitation for alignment evaluation: behavioral conformance under a finite test does not uniquely identify latent alignment when the system adapts to the evaluation context.

The key distinction is that, in alignment evaluation, the target of inference is not a fixed functional specification but a latent normative structure. As a result, even exhaustive testing within a regime may fail to distinguish between hypotheses that agree locally but diverge globally.

6.4 Relation to Inverse Reinforcement Learning

Inverse Reinforcement Learning (IRL) and related preference inference methods aim to recover latent objectives from observed behavior. Identifiability results in these frameworks typically rely on assumptions of stationarity, optimality, and policy independence from the data-collection process.

Our result does not contradict IRL-style inference. Rather, it delineates its domain of applicability. When agent behavior is evaluation-aware and interaction is finite, the assumptions required for IRL identifiability no longer hold. In this regime, behavioral evidence alone may be insufficient to uniquely identify latent objectives, even though such identification is possible in classical, non-reactive settings.

6.5 Synthesis

Taken together, these connections clarify the contribution of our analysis. While finite observation, observational equivalence, and non-identifiability are well-established concepts in inference theory, their interaction in the context of adaptive agents yields a distinct epistemic challenge. By formalizing behavioral alignment tests as epistemic instruments and allowing policies to condition on evaluation context, we show that alignment evaluation inherits structural limits analogous to those found in other domains of inverse inference, but amplified by endogeneity.

Our result should therefore be understood not as a rejection of empirical evaluation, but as a characterization of the conditions under which empirical evidence can and cannot support claims about latent alignment.

7 Modern Context: Evaluation Practice and the Limits of Behavioral Evidence

Contemporary alignment practice relies on increasingly sophisticated evaluation pipelines, ranging from multi-metric benchmarks to automated red-teaming [29, 38]. Despite their methodological diversity, these approaches share a common evidential structure: they infer properties of interest from finite interaction traces produced under a specified protocol. This section situates our non-identifiability result within this landscape, emphasizing two interacting constraints: (i) evaluation is necessarily finite, and (ii) the measurement instruments themselves may introduce additional noise, bias, or protocol dependence [27].

7.1 From General Capability Benchmarks to Alignment Evaluation

Large-scale efforts such as HELM [29] and GEM [17] have contributed to broader and more systematic assessment by combining multiple metrics and task families. However, such benchmarks remain finite